GATE CS&DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 9

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Recap of Previous Lecture

Topic

Topic

Topic

Topic

Topic

Predicate logic

Turn on Slide map

Topics to be Covered

- Topic Predicate logic
- Topic
- Topic
- Topic
- Topic

“Don’t let yesterday take up too much of today.”

Will Rogers

So Proposition wale valid

Rule of inference w.r.t Predicate Logic

- In predicate logic, rules of inference are used to derive conclusions from premises that contain quantifiers (like ∀ for "for all" and ∃ for "there exists"). These rules are extensions of propositional logic rules to handle predicates and quantifiers.
- Here are some key **rules of inference** in **predicate logic**:

Rule of inference w.r.t Predicate Logic

- 1. Universal Instantiation (UI):
- This rule allows us to derive a specific instance from a universally quantified statement.
- General Form: $\forall x\, P(x)$ implies $P(c)$
- If a property P(x) holds for all x, then it holds for any specific c.

$\forall x P(x) \longrightarrow P(c)$ will also be true

If for all x P(x) is true

C is any x in domain of x

Rule of inference w.r.t Predicate Logic

- Universal Generalization (UG):
- This rule allows us to generalize from a specific case to a universal statement.
- General Form: $P(c)$ implies $\forall x\, P(x)$
- If P(c) is true for an arbitrary c, then $P(x)$ is true for all x.

P(c) is valid for all 'c' it means $\forall x\, P(x)$

Rule of inference w.r.t Predicate Logic

- Existential Instantiation (EI):
- This rule allows us to introduce a specific instance from an existentially quantified statement.
- General Form: $\exists x\, P(x)$ implies $P(c)$
- If P(x) holds for some x, then it holds for a specific c, where c is a new symbol that doesn't appear anywhere else.

$\exists x P(x) \longrightarrow$ P(x) is true for some c

for some x, P(x) is true

Rule of inference w.r.t Predicate Logic

- Existential Generalization (EG):
- This rule allows us to conclude an existential statement from a specific instance.
- General Form: $P(c)$ implies $\exists x\, P(x)$
- If P(c) is true for a specific c, then there exists some x such that P(x) is true.

$\Rightarrow$ (If P(c) is true for any specific c $\longrightarrow$ $\exists_x P(x)$ is valid)

Rule of inference w.r.t Predicate Logic

- Existential Generalization (EG):
- This rule allows us to conclude an existential statement from a specific instance.
- General Form: $P(c)$ implies $\exists x\, P(x)$
- If P(c) is true for a specific c, then there exists some x such that P(x) is true.

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

So (LHS ≡ RHS)

$$\underbrace{\forall x\,(P(x) \wedge Q(x))}_{} \equiv \underbrace{\forall x\,(P(x))}_{} \wedge \underbrace{\forall x\,(Q(x))}_{}$$

- forall x [P(x) ∧ Q(x)]

Should be true

forall x P(x) and forall x Q(x) shd be True

shd be true

- we know that for all x P(x) ∧ Q(x) = 1, only if ∀x P(x), ∀x Q(x) = 1

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications (done)

$$\exists_x (P(x) \vee Q(x)) \equiv \underbrace{\exists_x (P(x))}_{\text{for some } x \; P(x)=\text{True}} \vee \underbrace{\exists_x (Q(x))}_{\text{for some } x \; Q(x)=\text{True}}$$

for some x $(P(x) \vee Q(x))$ is true

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

False

$$\forall_x \left(\overset{1}{P(x)} \vee \overset{0}{Q(x)} \right) \equiv \underbrace{\forall_x (P(x))}_{\text{for all } x \; P(x) \text{ is true}} \vee \underbrace{\forall_x (Q(x))}_{\text{for all } x \; Q(x) \text{ is true}}$$

$\Rightarrow$ for all x $\underbrace{(P(x) \vee Q(x))}_{\text{True}}$ is True

$(P(x)=1 \text{ OR } Q(x=1) \text{ OR } P(x)=Q(x)=1)$

OR

For RHS = True, either $P(x)=1$ for all x or $Q(x)=1$ for all x or both.

False $\Leftarrow$ $\overbrace{\forall x\,(P(x) \vee Q(x))}^{A} \longrightarrow \overbrace{\forall x\,(P(x)) \vee \forall x\,(Q(x))}^{B}$ $\Rightarrow$ if A is true then B is true

$\Rightarrow \underbrace{\forall x\,(P(x)) \vee \forall x\,(Q(x))}_{A} \longrightarrow \underbrace{\forall x\,(P(x) \vee Q(x))}_{B}$ $\Rightarrow$ True

Here if A is true then B is also true

P	Q
1	0
1	1
1	0

P	Q
0	1
1	1
1	1

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

not equal

$$\exists_x (P(x) \wedge Q(x)) \equiv \underbrace{\exists_x (P(x))}_{\text{for some } x \; P(x)=\text{True}} \wedge \underbrace{\exists_x (Q(x))}_{\text{for some } x \; Q(x)=\text{True}}$$

for at least one x $(P(x) \wedge Q(x))$ is true

→ True only if $P(x)=Q(x)=1$ exactly for same x values.

So if LHS is true thus $P(x)$ and $Q(x)$ are true for same x values (for few only)

Now let $P(x), Q(x)$ are true for some x

RHS = True, it does assume LHS = True

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

not equal

$$\underbrace{\exists x\,(P(x) \wedge Q(x))}_{A} \qquad \underbrace{\exists x\,(P(x)) \wedge \exists x\,(Q(x))}_{B}$$

$\Rightarrow$ $A \rightarrow B$ if A is true then B is true

$\times$ $\cancel{B \rightarrow A}$ if B is true then A is true

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

$\forall_x (P(x) \rightarrow Q(x)) \equiv \underbrace{\forall_x (P(x))} \rightarrow \underbrace{\forall_x (Q(x))}$ ⇗ Fail

- For all x $(P(x) \rightarrow Q(x))$

  is true, if P(x) is false or P(x) and Q(x) both are true

  If LHS is true ⇒ i.e $P(x) \rightarrow Q(x)$ is true for all x.

RHS will be true if

$\forall x (P(x)) = 0$ or false

OR $\forall x (P(x)) = 1$ and $\forall x (Q(x) = 1)$

$(1 \rightarrow 1) = 1$

$0 \rightarrow 1 = 1$

P	Q
1	1
1	1
1	1

P	Q
1	1
0	1
1	1

P	Q
1	1
0	0
1	1

P	Q
1	1
1	0
1	1

$0 \rightarrow 0$ True

$1 \rightarrow 0 = 0$ false

$\forall x\ (P(x) \rightarrow Q(x))$

So this will be true if for all x $P(x) \rightarrow Q(x)$ is true

$P(x)$	$Q(x)$
0	0
0	1
1	1
1	1

$P(Q)$	$Q(x)$
1	1
1	1
1	1

$P(Q)$	$Q(x)$
0	1
0	0
0	1
0	0
0	1
0	0

P	Q
1	0
1	0
1	0
1	0

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications ⇗ fail

$$\underbrace{\forall_x (P(x) \rightarrow Q(x))}_{A} \equiv \underbrace{\forall_x (P(x)) \rightarrow \forall_x (Q(x))}_{B}$$

If A is true then B is true.

$A \rightarrow B$

If B is true then A is not sure

$\cancel{B \rightarrow A}$

$(1 \rightarrow 1) = 1$

$0 \rightarrow 1 = 1$

P	Q
1 ✓	1
1	1
1	1

P	Q
1	1
0 ✓	1
1	1

P	Q
1 ✓	1
0	0
1	1

$\Downarrow$

$0 \rightarrow 0$

True

P	Q
1	1
1	0
1	1

$\Downarrow$

$1 \rightarrow 0 = 0$

False

P	Q
1	0
0	1

$\downarrow$

$0 \rightarrow 0$

[NAT]

1. $\forall x[P(x) \wedge Q(x)] \equiv [\forall x\, P(x)] \wedge [\forall x\, Q(x)]$

2. $\exists\, x\, [P(x) \vee Q(x)] \equiv \exists_x P(x) \vee \exists_x Q(x)$

3. $[\forall\, x\, P(x)] \vee [\forall\, x\, Q(x)] \Rightarrow \forall\, x\, [P(x) \vee Q(x)]$

4. $\exists\, x\, [P(x) \wedge Q(x)] \Rightarrow [\exists\, x\, P(x)] \wedge [\exists\, x\, Q(x)]$

5. $\forall\, x\, [P(x) \rightarrow Q(x)] \Rightarrow [\forall\, x\, P(x)] \rightarrow [\forall\, x\, Q(x)]$

done ✓

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$
4. $\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$
5. $\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$
6. $\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$
7. $\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$
8. $\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

Q is Independent of $x \Rightarrow$

LHS is true

$\forall x(P(x) \wedge Q) \equiv$ (for all x $P(x)$=True And Q=True)

$\Rightarrow$ for all x $P(x)$=True, Q shd be true

$\exists x(P(x) \vee Q) \equiv$ Some x $P(x)$=True OR Q=True

$\equiv \exists x\, P(x) \vee Q$

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\boxed{\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q}$
4. $\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$
5. $\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$
6. $\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$
7. $\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$
8. $\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

- $\forall x\, (P(x) \vee Q)$

$\boxed{\text{if } Q = \text{True}}$

$\boxed{\text{if } Q = \text{False}}$

$(\forall x\, P(x)) \vee Q$

Then both are True always

$\forall x\, (P(x)) \equiv \forall x\, (P(x))$

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$
4. $\boxed{\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q}$
5. $\boxed{\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)}$
6. $\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$
7. $\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$
8. $\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

$\exists_x[P(x) \wedge Q]$ $\qquad$ $\exists_x P(x) \wedge Q$

$$\left\{\begin{array}{ll} Q: \text{True} & \exists_x(P(x)) \equiv \exists_x(P(x)) \\ Q: \text{False} & \text{False} \equiv \text{False} \end{array}\right\}$$

$$\boxed{\forall_x(P \rightarrow Q(x)) \equiv P \rightarrow \forall_x Q(x)}$$

$$\boxed{(A \to B \equiv \neg A \vee B)}$$

So $\forall x\,(P \to Q(x))$

$$\equiv \forall x\,(\sim P \vee Q(x))$$

$$\equiv \sim P \vee \forall x\, Q(x))$$

$$\equiv P \to \forall x\, Q(x)$$

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$
4. $\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$
5. $\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$
6. $\boxed{\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)}$ $\Longrightarrow$ $\exists x\,(P \rightarrow Q(x)) \equiv \exists x\,(\neg P \vee Q(x))$

   $\equiv \neg P \vee \exists x\, Q(x)$

   $\equiv P \rightarrow \exists x\, Q(x)$
7. $\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$
8. $\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$
4. $\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$
5. $\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$
6. $\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$
7. $\boxed{\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q}$
8. $\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

$(\neg A \vee B \equiv A \rightarrow B)$

$\forall x(P(x) \rightarrow Q)$

$\equiv \forall x(\neg P(x) \vee Q)$

$\equiv \forall x\, \neg P(x) \vee Q$

$\equiv \neg(\forall x\, \neg P(x)) \rightarrow Q$

$\equiv (\exists x\, P(x) \rightarrow Q)$

[NAT]

1. $\forall x[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$
2. $\exists x\, [P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$
3. $\forall x\, [P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$
4. $\exists x\, [P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$
5. $\forall x[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$
6. $\exists x\, [P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$
7. $\forall x\, [P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$
8. $\boxed{\exists x\, [P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q}$

$(\neg A \vee B \equiv A \rightarrow B)$

$$\exists x\, [P(x) \rightarrow Q] \equiv \exists x\, (\neg P(x) \vee Q)$$
$$\equiv (\exists x\, \neg P(x) \vee Q)$$
$$\equiv \neg \exists x\, \neg P(x) \rightarrow Q$$
$$\equiv (\forall x\, P(x) \rightarrow Q)$$

[MCQ]

Let P(x) and Q(x) be arbitrary predicates. Which of the following statements is always TRUE?

A. $(\forall x\,(P(x) \vee Q(x))) \Rightarrow ((\forall x\,P(x)) \vee (\forall x\,Q(x)))$

B. $(\forall x\,(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x\,P(x)) \Rightarrow (\forall x\,Q(x)))$

C. $(\forall x\,(P(x) \Rightarrow \forall x\,Q(x))) \Rightarrow (\forall x\,(P(x) \Rightarrow Q(x)))$

D. $((\forall x\,(P(x)) \Leftrightarrow (\forall x\,Q(x))) \Rightarrow (\forall x\,(P(x) \Leftrightarrow Q(x)))$

So if A is true then B is not surely true

$$\overbrace{\left[\forall x\,(P(x)) \equiv \forall x\, Q(x)\right]}^{A} \Longrightarrow \underbrace{\forall x\,\left(P(x) \equiv Q(x)\right)}_{B}$$

- if A is true then it implies B is true

equivalence is true

LHS True $\Rightarrow$ $\forall x P(x) = 1$, $\forall x Q(x) = 1$
i.e at all place $P(x)$ and $Q(x) = 1$ $\Rightarrow$ thus RHS is true

LHS True $\Rightarrow$ $\forall x P(x) = 0 = \forall x Q(x) = 0$
thus at all place $P(x)$ and $Q(x)$ are not 1 $\Rightarrow$ $\underbrace{\forall x\,\left(P(x) \equiv Q(x)\right)}_{\text{false}}$

P	Q
1	0
0	1

$$\underbrace{\forall x\,(P(x) \equiv Q(x))}_{\text{For all } x\ P(x)=Q(x)} \Longrightarrow \left(\forall x\,(P(x)) \equiv \forall x\,(Q(x))\right)$$

True.

P	Q
0	0
0	0
0	0
⋮	⋮

P	Q
0	0
0	0
0	0
0	0

P	Q
1	1
1	1
1	1
1	1

[MCQ]

Which of the following predicate calculus statements is/are valid

A. $(\forall x)\ P(x) \vee (\forall x)\ Q(x) \rightarrow (\forall x)\ \{P(x) \vee Q(x)\}$

B. $(\exists x)\ P(x) \wedge (\exists x)\ Q(x) \rightarrow (\exists x)\ \{P(x) \wedge Q(x)\}$

C. $(\exists x)\ \{P(x) \vee Q(x)\} \rightarrow (\forall x)\ P(x) \vee (\forall x)\ Q(x)\}$

D. $(\exists x)\ \{P(x) \vee Q(x)\} \rightarrow (\exists x)\ P(x) \vee \forall x\ Q(x)\}$

[NAT]

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\, P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\, P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\, P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\, P(x) \wedge Q$

5. $\forall x\,[P \rightarrow Q(x)] \equiv P \rightarrow \forall x\, Q(x)$

6. $\exists x\,[P \rightarrow Q(x)] \equiv P \rightarrow \exists x\, Q(x)$

7. $\forall x\,[P(x) \rightarrow Q] \equiv \exists x\, P(x) \rightarrow Q$

8. $\exists x\,[P(x) \rightarrow Q] \equiv \forall x\, P(x) \rightarrow Q$

[MCQ]

Which of the following predicate calculus statements is/are valid

A. $\forall(x)\ (P(x) \Rightarrow Q(x)) \Rightarrow ((\forall(x)\ P(x)) \Rightarrow (\forall(x)\ Q(x)))$

B. $\exists\ x\ (P(x) \vee Q(x))) \Rightarrow ((\exists\ x\ P(x)) \Rightarrow (\exists\ x\ Q(x)))$

C. $\exists\ x\ (P(x) \wedge Q(x)) \Leftrightarrow ((\exists\ x\ P(x)) \wedge (\exists\ x\ Q(x)))$

D. $\forall(x)\ \exists y\ P(x, y) \Rightarrow \exists\ y\ \forall(x)\ P(x\ , y)$

[MCQ]

Let P(x) and Q(x) be arbitrary predicates. Which of the following statements is always TRUE?

A. $(\forall x\,(P(x) \vee Q(x))) \Rightarrow ((\forall x\,P(x)) \vee (\forall x\,Q(x)))$

B. $(\forall x\,(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x\,P(x)) \Rightarrow (\forall x\,Q(x)))$

C. $(\forall x\,(P(x) \Rightarrow \forall x\,Q(x))) \Rightarrow (\forall x\,(P(x) \Rightarrow Q(x)))$

D. $((\forall x\,(P(x)) \Leftrightarrow (\forall x\,Q(x))) \Rightarrow (\forall x\,(P(x) \Leftrightarrow Q(x)))$

[MCQ]

Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x.

A. $\exists x (P(x) \wedge W) \equiv (\exists x P(x)) \wedge W$

B. $\forall x (P(x) \rightarrow W) \equiv (\forall x P(x)) \rightarrow W$

C. $\exists x (P(x) \rightarrow W) \equiv (\forall x P(x)) \rightarrow W$

D. $\forall x (P(x) \vee W) \equiv (\forall x P(x)) \vee W$

[MCQ]

Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.

A. $[\beta\ (\exists\ x, \alpha(x))] \rightarrow [\forall\ x, \beta - \alpha(x)]$

B. $[\exists\ x, \beta \rightarrow (\alpha(x))] \rightarrow [\beta \rightarrow (\forall\ x, \alpha(x))]$

C. $[\exists\ x, (\alpha(x)) \rightarrow \beta] \rightarrow [\forall\ x, \alpha(x) \rightarrow \beta]$

D. $[\forall\ x, (\alpha(x)) \rightarrow \beta] \rightarrow [\forall\ x, \alpha(x) \rightarrow \beta]$

[MCQ]

Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

A. $((\forall x)\ [\alpha] \Rightarrow (\forall\ x)\ [\beta]) \Rightarrow (\forall x)\ [\alpha \Rightarrow \beta])]$

B. $((\forall x)\ [\alpha] \Rightarrow (\exists\ x)\ [\alpha \wedge \beta])$

C. $(\forall x)\ [\alpha \vee \beta] \Rightarrow (\exists\ x)\ [\alpha] \Rightarrow (\forall x)\ [\alpha]$

D. $((\forall x)\ [\alpha \Rightarrow \beta] \Rightarrow ((\forall\ x)\ [\alpha]) \Rightarrow (\forall x)\ [\beta])$

THANK - YOU